Year 13 Maths - Functions

Similar questions - Modulus functions and one-one functions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $\mathrm{f}(x) = |2x-4|+1$ for all real x.

(a) Sketch the graph of $y = \mathrm{f}(x)$, giving the coordinates of the vertex and of the point where it crosses the y-axis. **[3]**

(b) Solve $\mathrm{f}(x) = 7$. **[3]**

(c) Solve $\mathrm{f}(x) > x$. **[4]**

(d) Explain why f has no inverse, and state a domain on which it does. **[2]**

2 $\mathrm{h}(x) = x^2 - 4x$ for all real x.

(a) Show that h is a many-one function, by finding two values of x with the same image. **[2]**

(b) State the largest domain of the form $x \geqslant k$ for which h is one-one, and the range of h on this domain. **[2]**

(c) On this domain, find $\mathrm{h}^{-1}(x)$, stating its domain. **[3]**

(d) Find the value of $\mathrm{h}^{-1}(5)$ and verify it using h. **[2]**

Question 1

(a) V-shaped graph, vertex $(2,1)$ (where $2x-4=0$), crossing the y-axis at $(0,5)$. Gradient ∓ 2 either side

(b) $|2x-4|=6$, so $2x-4=\pm 6$ and $x=5$ or $x=-1$

(c) For $x \geqslant 2$: $2x-3>x$ gives $x>3$. For $x<2$: $5-2x>x$ gives $x<\frac{5}{3}$. So $x<\frac{5}{3}$ or $x>3$

(d) f is many-one (e.g. $\mathrm{f}(0)=\mathrm{f}(4)=5$). It has an inverse on $x \geqslant 2$ (or $x \leqslant 2$)

Question 2

(a) $\mathrm{h}(1)=-3$ and $\mathrm{h}(3)=-3$, so two inputs give the same output

(b) $\mathrm{h}(x)=(x-2)^2-4$, vertex at $x=2$: $x \geqslant 2$, range $\mathrm{h}(x) \geqslant -4$

(c) $y=(x-2)^2-4$, so $x=2+\sqrt{y+4}$. $\mathrm{h}^{-1}(x)=2+\sqrt{x+4}$, domain $x \geqslant -4$

(d) $\mathrm{h}^{-1}(5)=2+\sqrt{9}=5$. Check: $\mathrm{h}(5)=25-20=5$